Audio Multi-View Spoofing Detection Framework Based on Audio-Text-Emotion Correlations

Junyan Wu, Qilin Yin, Ziqi Sheng, Wei Lu, *Member, IEEE*, Jiwu Huang, *Fellow, IEEE*, and Bin Li, *Senior Member, IEEE*

***Abstract*— In recent years, audio spoofing detection has received widespread attention for protecting personal privacy and social security. Despite the significant progress achieved in audio single-view spoofing detection, challenges remain with regard to addressing unknown spoofing attacks in realistic scenarios. To solve these challenging problems, in this paper, we introduce a novel audio multi-view spoofing detection framework (AMSDF), whose goal is to capture both intra-view and inter-view cues by measuring correlations within audio multi-view features (i.e., audio-emotion-text) for audio spoofing detection. In general, different view features are inherently interconnected in the real patterns, while they may present unnatural correlations in the spoofing patterns. Therefore, more discriminative cues can be mined by utilizing their complex interactions, which is beneficial to the audio spoofing detection task. To this end, an intra-view graph attention mechanism (IGAM) is first utilized to aggregate each intra-view node within the same view. Subsequently, a heterogeneous graph fusion module (HGFM) is applied to measure correlations within inter-view nodes, which are enhanced with a master node for comprehensive analysis purposes. Finally, a group-based readout scheme (GRS) is designed to capture and preserve the most distinctive cues by leveraging the strengths of different feature sets, thereby effectively distinguishing subtle differences between real and spoofing audio. The experimental results show that our proposed framework can achieve better performance than that of the state-of-the-art methods, especially in realistic scenarios. The code and pre-trained models are available at https://github.com/ItzJuny/AMSDF.**

***Index Terms*— Multimedia forensics, audio spoofing detection, multi-view learning, graph attention mechanism.**

I. Introduction

THE rapid advances in audio deepfake technology have made lifelike forgeries more accessible to the general public. Driven by massive amounts of data and powerful deep neural network (DNN) variants, i.e., generative adversarial networks (GANs) [1], [2], [3], flow-based models [4], [5], [6] and diffusion models [7], [8], [9], the existing audio deepfake methods [10], [11], [12], [13], [14], [15] can synthesize realistic speech that is indistinguishable from human speech. The dangers of such technologies are obvious, as malicious users can easily counterfeit someone's identity by falsifying the speech contents and voice characteristics even without consent. Currently, numerous telemarketing fraud incidents and the widespread dissemination of spoofing audio through the internet are causing significant damage to individuals, organizations, and society. Therefore, developing spoofing detection systems that can effectively detect audio spoofing attacks is vital.

To meet this challenge, researchers have dedicated their efforts to developing audio spoofing detection methods in recent years. Front-end feature extractors based on the pre-transformations of raw waveform, coupled with a back-end machine learning classifier (such as a support vector machine or a Gaussian mixture model) or a DNN-based decoder have become routine solutions for audio spoofing detection [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34]. However, the reliance on hand-crafted features may not be able to encompass all forgery factors, potentially not representing the best detection solution. As a result, researchers have proposed several end-to-end detection methods [35], [36], [37], [38], [39] that operate directly on the raw waveform to automatically capture potential forgery artifacts. Such methods automatically learn spoofing cues from raw waveform, which can be further improved by adding forgery hints provided by hand-crafted features [40], [41]. In addition, several current studies have explored features different from the above approaches to mine forgery artifacts arising from the defects of audio-synthesized generators. These findings indicate that using a pre-trained automatic speech recognition (ASR) model [42], [43], [44] or a speech emotion recognition (SER) model [45] along with a back-end classifier can offer a forward-looking solution for audio spoofing detection. While promising results have been shown in audio single-view spoofing detection, the challenge remains to address unknown attacks in realistic scenarios.

There are two main limitations in the current audio spoofing detection methods. Firstly, most existing methods rely solely on single-view features, which may not be sufficient to uncover multi-variety spoofing cues, resulting in detection difficulties in realistic scenarios. Secondly, current methods focus on detecting specific forgery artifacts in the temporal and frequency domains, which may not generalize well to

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Junyan Wu, Qilin Yin, Ziqi Sheng, and Wei Lu are with the School of Computer Science and Engineering, Ministry of Education Key Laboratory of Information Technology, Guangdong Province Key Laboratory of Information Security Technology, Sun Yat-sen University, Guangzhou 510006, China (e-mail: wujy298@mail2.sysu.edu.cn; yinqlin@mail2.sysu.edu.cn; shengzq@mail2.sysu.edu.cn; luwei3@mail.sysu.edu.cn).

Jiwu Huang is with Guangdong Laboratory of Machine Perception and Intelligent Computing, Faculty of Engineering, Shenzhen MSU-BIT University, Shenzhen 518116, China (e-mail: jwhuang@smbu.edu.cn).

Bin Li is with Guangdong Key Laboratory of Intelligent Information Processing and Shenzhen Key Laboratory of Media Security, Shenzhen University, Shenzhen 518060, China (e-mail: libin@szu.edu.cn).

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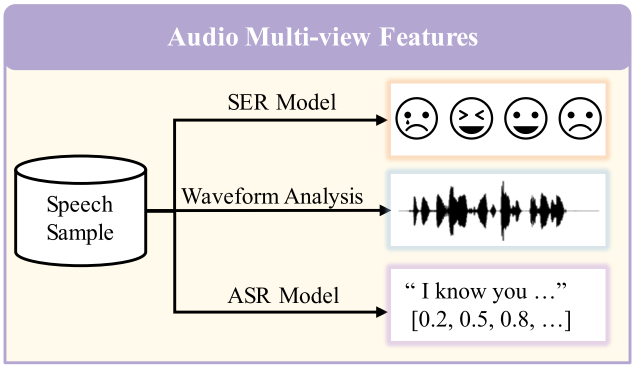


Fig. 1. Multi-view information via speech processing — Automatic Speech Recognition (ASR), waveform analysis, and Speech Emotion Recognition (SER).

unseen spoofing attacks that introduce novel forgery artifacts. Additionally, in compressed audio scenarios, forgery artifacts may be distorted, making detection challenging. Therefore, we are motivated to seek more generalized and robust spoofing cues to detect unknown audio spoofing attacks from a variety of sources in realistic scenarios.

To address the above challenges, we utilize a multi-view learning approach in this paper. Similar to multi-model learning, which involves integrating different types of data [46], [47], [48] for analysis, multi-view learning is also well-recognized for its ability to utilize diverse feature perspectives derived from the same input to tackle difficult tasks. In the exploration of multi-view learning for spoofing detection, studies such as [49] and [50] have demonstrated improvements in the generalization and robustness performance of detecting spoofing faces and manipulated images. However, it has not been widely adopted in the audio spoofing detection task, where the existing methods mainly focus on capturing forgery artifacts based on audio single-view features. As depicted in Fig. 1, multiple pipelines are available for parsing audio information. In general, audio can relay emotional cues (the SER pipeline), such as happiness and sadness, and can also convey the content spoken by a person (through the ASR pipeline). Human speech is generally perceived as more personable and engaging than synthesized speech, as spoofing algorithms may struggle to accurately integrate emotional nuances with the corresponding textual content. Additionally, the alignment process used in synthesized generators may result in strong audio-text correlations among spoofing patterns. That is to say, these multi-view features are inherently interconnected in the real patterns, while unnatural correlations within the spoofing patterns tend to be presented across different attacks. Inspired by this observation, the main motivation of our paper involves modeling spoofing cues by identifying discrepancies in the correlations among multi-view features that are naturally aligned in real audio but are distorted in spoofing audio.

Therefore, we are motivated to propose a novel audio multi-view spoofing detection framework (AMSDF) based on audio-text-emotion correlations. In this framework, we first utilize multiple feature extractors to derive multi-view information from audio, including audio-view features learned from raw waveform, emotional semantics obtained from an SER system, and textual representations acquired from an ASR system. Instead of capturing specific forgery artifacts, this framework is learned to mine unnatural audio-text-emotion correlations inherent in spoofing patterns to provide generalized and robust forgery cues for detection. To this end, we design the multi-view correlation measurement network to capture both intra-view and inter-view forgery cues. This network incorporates an intra-view graph attention mechanism (IGAM) and a heterogeneous graph fusion module (HGFM), facilitating comprehensive correlation analysis of the observed audio spoofing patterns. Furthermore, to fully exploit forgery cues derived from diverse feature sets, we introduce the group-based readout scheme (GRS) in the back-end classifier. The experimental results show that our proposed framework can achieve better performance than that of the state-of-the-art methods, especially in realistic scenarios. The key contributions of this paper are summarized as follows:

- A novel multi-view learning framework named AMSDF is proposed to analyze complex interactions among audio-text-emotion features, which can effectively counter diverse spoofing attacks in realistic scenarios.
- The IGAM and the HGFM are specifically utilized to model high-level spoofing cues within intra-view and inter-view correlations, respectively. These components could benefit the identification of unnatural correlations inherent in spoofing patterns.
- The GRS is employed to enhance discriminate features by leveraging the strengths of different forgery cues, thereby effectively distinguishing real and spoofing audio.

The remainder of this paper is organized as follows. Section II discusses the related works. Then, Section III describes the proposed framework. The experimental results are analyzed in Section IV. Finally, Section V presents the conclusion, including the shortcomings of the proposed framework and future work ideas.

II. Related Works

A. Multi-View Spoofing Detection

In recent years, the multi-view spoofing detection task has gained significant attention, with the aim of fully exploiting diverse feature perspectives from the input data to learn better representations. For example, Kuang et al. [49] acquired multi-view features from face images, including RGB, near-infrared (NIR) spectra, and depth data. This approach aided in the fusion of these features by the design of a multi-view weight-adaptive block, thereby improving the face spoofing detection performance. Additionally, Chen et al. [50] jointly exploited the image noise distribution and boundary artifacts surrounding tampered regions to address the image manipulation detection task. The remarkable results achieved in both face spoofing and image manipulation detection tasks have demonstrated the effectiveness of multi-view learning. Unfortunately, most of the existing audio spoofing detection methods focus on designing audio single-view features, which motivates us to include more feature perspectives within audio to improve detection performance.

B. Graph Attention Networks in Audio Spoofing Detection

The objective of audio spoofing detection involves determining whether a given speech sample originates from a natural person (referred to as real speech) or is synthesized by a spoofing algorithm (referred to as fake speech). Graph attention networks (GATs) have proven to be powerful in modeling relationships within audio-view features, providing advanced detection solutions. For example, Hemlata et al. [39] utilized individual temporal GAT (GAT-T) and spectral GAT (GAT-S) to model the relationships between adjacent frames and distinct sub-bands, respectively. Expanding upon this research, Tak et al. [37] introduced the RawGAT-ST structure, incorporating spectro-temporal attention to further model the relationships between the spectral and temporal domains. Moreover, Jung et al. [38] proposed the audio anti-spoofing method using integrated spectro-temporal (AASIST) GATs that capture spoofing cues spanning diverse temporal and spectral intervals in a heterogeneous way. In summary, audio spoofing detection methods based on GATs have achieved promising intra-dataset evaluation performance, i.e., both training and testing on the same dataset. However, a noticeable decline in their performance can be observed during cross-dataset or robust-dataset evaluation. This decline may be attributed to the assumption that these spectro-temporal artifacts rely on the single audio-view and may overfitted to specific spoofing attacks that are different from unknown spoofing attacks. Thus, we are motivated to utilize the anti-spoofing-based GATs [37], [38], which have the advantage of correlation modeling, to mine audio-text-emotion correlations unnaturally aligned in spoofing audio for detecting.

C. Data Augmentation Methods in Audio Spoofing Detection

Various data augmentation strategies have been adopted to enhance model robustness in the audio spoofing detection task. For example, Tak et al. [51] introduced the raw data boosting and augmentation method (RawBoost), specifically tailored for the ASVS2021 datasets. This study involved detailed experiments on combinations of linear and non-linear convolutive noise, impulsive signal-dependent additive noise, and stationary signal-independent additive noise. Similarly, diverse augmentation techniques like mix-up methods, finite impulse response (FIR) filters, room impulse responses (RIR), and noise enhancement scheme were also studied in [23]. These data augmentation methods have been shown to be effective at boosting robust performance. However, this paper takes a different approach that captures high-level spoofing cues by analyzing correlations among different views to enhance the robustness of the framework. Previous works like [43], [44], and [42] utilized textual-view features obtained from a large-scale ASR pre-trained model, and Conti et al. [45] have explored emotional-view features. While these methods have made significant improvements by utilizing features from other views, they rely on textual or emotional features to capture specific forgery artifacts and remain single-view detection solutions. Therefore, we introduce a novel audio multi-view spoofing detection framework to fully exploit multi-view features from the audio, textual, and emotional views. Unlike existing works, our framework can integrate spoofing cues derived from multi-view correlations via a designed multi-view correlation measurement network, which presents a forward-looking solution for audio spoofing detection.

III. Proposed Framework

In this section, we first introduce an overview of our proposed framework. Then, we will illustrate key components, including the extraction of multi-view features and the construction of the multi-view correlation measurement network and back-end classifier.

A. Overview

Since audio spoofing attacks continue to evolve, unknown forgery artifacts, encoding, and transmission introduced in realistic spoofing scenarios are challenging existing detection methods. Thus, we seek to capture generalized and robust spoofing cues by mining inherent defects introduced by spoofing patterns, i.e., undermined natural correlations among audio multi-views. Based on the multi-view learning idea, as shown in Fig. 2, the proposed framework consists of three main components: front-end feature extractors, a multi-view correlation measurement network, and a back-end synthesized classifier. First, to acquire multi-view information, the front-end extractors process the input speech into audio-, emotional-, and textual-view feature sets as $\mathcal{F} = \{\mathcal{F}_A, \mathcal{F}_E, \mathcal{F}_T\}$.

Second, a multi-view correlation measurement network is proposed to capture both intra-view and inter-view forgery correlation cues. During the HGFM operations, multi-view features are adapted to each other and make the discrepancies in the multi-view correlations between real and spoofing audio better discerned. Specifically, each node derived from $\mathcal{F}$ is connected to all other nodes in the same view. Consequently, by applying separate IGAMs to the multi-view features, the strength of the relationships is adjusted to model the intra-view spoofing cues, resulting in sets of homogeneous graphs $\mathcal{G}$. Then, the correlations between the pairs of correlated graphs within specific relationships are measured by the HGFM, resulting in sets of heterogeneous graphs $\mathcal{H}$. Moreover, the aggregated information is enhanced by the interaction of the heterogeneous nodes with master nodes $\mathcal{M}$ from a global view.

Finally, to fully exploit spoofing correlation cues under multiple relationships, a multi-view synthesized classifier is designed. Different spoofing cues represented by $\mathcal{U} = \{\mathcal{G}, \mathcal{H}, \mathcal{M}\}$ are divided into K ($K = 3$) groups. A GRS is applied to each group to preserve the crucial forgery information, resulting in the corresponding group-level output $\mathcal{U}_k^P$. The last hidden states, denoted as O, are obtained by concatenating these outputs, and then are fed into a multi-layer perceptron (MLP) to generate a binary prediction Y_{pred}. Furthermore, detailed elaborations of the above structures are presented in the following sections.

B. Multi-View Feature Extractors

It is widely recognized that audio conveys rich information, such as the textual content and emotional nuances of

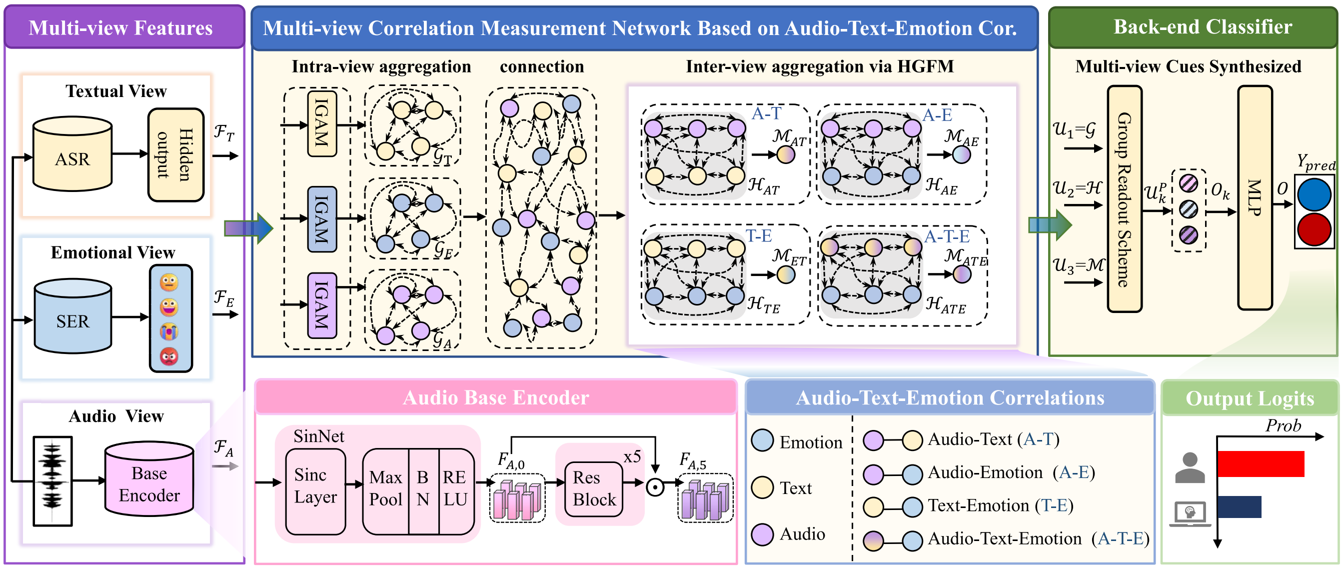


Fig. 2. The structure of the proposed multi-view learning framework for audio spoofing detection. This framework extracts multi-view features from raw waveform, an SER system, and an ASR system. It then derives intra-view and inter-view spoofing cues using a multi-view correlation measurement network and preserves the most discriminative cues through the group-based readout scheme in the classifier.

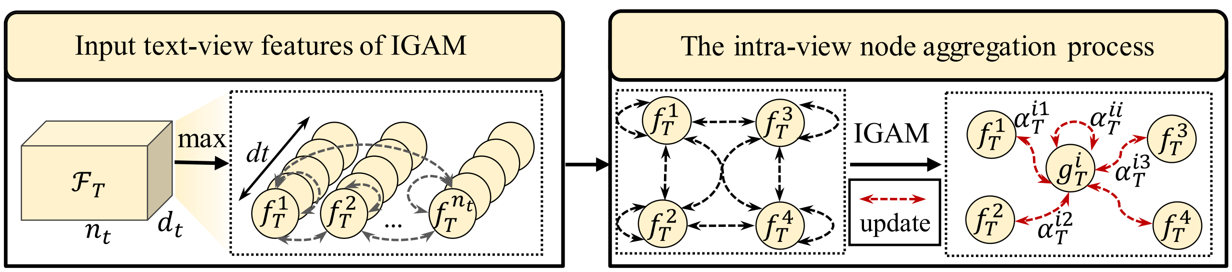


Fig. 3. Example of the intra-view graph attention mechanism employed for the textual-view features. The left side shows the process of constructing the input feature map $\mathcal{F}_T$. The right side displays the aggregation process from node $f_T^{(i)}$ to node $g_T^{(i)}$ via the IGAM.

a speaker, whose correlations are naturally aligned in real patterns and are disrupted in spoofing patterns. Therefore, the correlation analysis among audio multi-view information benefits audio spoofing detection as it aims to model the high-level spoofing cues inherent in diverse spoofing attacks. To acquire multi-view information, the proposed framework initially incorporates feature extractors to capture audio, emotional, and textual features from pre-trained models. Specifically, the extraction of audio-view features $\mathcal{F}_A$ is based on the base encoder designed in advanced solutions, like [36], [37], and [38]. This audio encoder comprises a Sinc layer and various residual blocks. Research has proven that such audio-view features can be learned to model spoofing cues from raw waveform directly. Additionally, the pre-trained ASR model, namely "wav2vec 2.0 XLS-R 300M" [52], is adopted because powerful representations learned from extensive speech corpora can boost detection performance [43], [44]. The textual-view features $\mathcal{F}_T$, are derived from the last hidden states of the ASR encoder and then fine-tuned within the whole framework. It is demonstrated that with fine-tuning, the downstream model can achieve state-of-the-art performance in tasks such as speech translation, speech recognition, and language identification. Moreover, previous studies [45], [53], [54] have indicated that emotional cues can be used to differentiate between real and spoofing speech, as current audio synthesized algorithms struggle to effectively incorporate natural emotional nuances into spoofing speech. The utterance-level emotional features $\mathcal{F}_E$ are extracted from the last hidden layer (preceding the final fully connected layer) of an SER model [45]. Consistent with the procedure in [45], this model is first pre-trained on the IEMOCAP dataset, and its emotion recognition capabilities are then tailored for spoofing detection via fine-tuning.

C. Multi-View Correlation Measurement Network

Graph attention networks are known for their ability to model correlation in various relationships. Unlike existing works that utilize GATs to model the relationship in the audio spectro-temporal domain, the multi-view correlation measurement network based on GATs is designed to model intra- and inter-view relationships within audio-text-emotion view, which is the core of our proposed AMSDF. As illustrated in Fig. 2, this network mainly comprises the IGAM and the HGFM to dynamically assign importance scores to nodes and influence the graph aggregation process. Let $v \in \{A, E, T\}$ denote the view in the multi-view set, including audio-view (A), emotional-view (E), and textual-view (T). First, node aggregations in the $\mathcal{F}_v$ are facilitated by an IGAM that learns weights reflecting the strength of the relationships between a node itself and its neighbors. As shown in Fig.3, each node is connected to all other nodes without any set directions in the same view. Therefore, the v-view features $\mathcal{F}_v \in \mathbb{R}^{n_v \times d_v}$ is used to create an initially graph that contains n_v nodes, each with d_v dimension. Specifically, the attention weight e_v^{ij}, which indicates the strength of the relationship between the i-th node $f_v^{(i)}$ and the j-th node $f_v^{(j)}$ is obtained by means of:

$$e_v^{ij} = W_v \cdot \left(f_v^{(i)} \odot f_v^{(j)} \right), \tag{1}$$

where $\odot$ signifies the element-wise multiplication, and $W_v \in \mathbb{R}^{1 \times d_v}$ is a learnable map for adjusting the intra-view relationship weights via a dot product. This weight is then normalized using the softmax function, resulting in

$$\alpha_v^{ij} = \mathrm{softmax}_j \left(e_v^{ij} \right) = \frac{\exp\left(e_v^{ij} / \tau \right)}{\sum_{l \in \mathcal{N}(f_v^{(i)})} \exp\left(e_v^{il} / \tau \right)}, \tag{2}$$

where τ is a temperature parameter that adjusts the sharpness of the softmax distribution and $\mathcal{N}(f_v^{(i)})$ represents the set of neighboring nodes of node i, including itself. Subsequently, the aggregated information of node i is calculated by the weighted sum of its connected nodes, which is described as follows:

$$f_v^{(i)'} = \sum_{j \in \mathcal{N}(f_v^{(i)})} \alpha_v^{ij} f_v^{(j)}, \tag{3}$$

where α_v^{ij} is the normalized attention weight between nodes i and j. Finally, node i is updated through a dual projection operation that effectively leverages the aggregated information while retaining the crucial cues derived from the original features:

$$g_v^{(i)} = \mathrm{SELU}\left(W_{agg} \left(f_v^{(i)'} \right) + W_{res} \left(f_v^{(i)} \right) \right), \tag{4}$$

where $\mathrm{SELU}(\cdot)$ is the activation function, the weight matrix W_{agg} is used to transform the nodes containing the intra-view aggregation information, and W_{res} is applied to project the original nodes. In this context, $\mathcal{G}_v = \left\{ g_v^{(1)}, g_v^{(2)}, \ldots, g_v^{(n_v)} \right\}$ with $g_v^{(i)} \in \mathbb{R}^{d_v}$ is derived, representing an updated homogeneous graph that models the forgery correlation cues residing in the intra-view nodes. Then, the HGFM designed in [38] is introduced to measure the correlation among pairs of correlated graphs under various relationships. The considered inter-view correlations are described as follows:

Audio-Text (AT) correlation: This correlation evaluates the consistency between audio and its textual transcriptions, distinguishing between artificially matched content in spoofing audio and genuine alignment in real audio.

Audio-Emotion (AE) correlation: This correlation assesses the relationship between the emotional tone of audio and its expected emotional context, which helps in detecting spoofing audio that lacks natural emotional consistency.

Emotion-Text (ET) correlation: This correlation examines the alignment between the emotional content of the text and the associated emotions. It aids in identifying artifacts in spoofing patterns where the emotional cues may not match the given textual content.

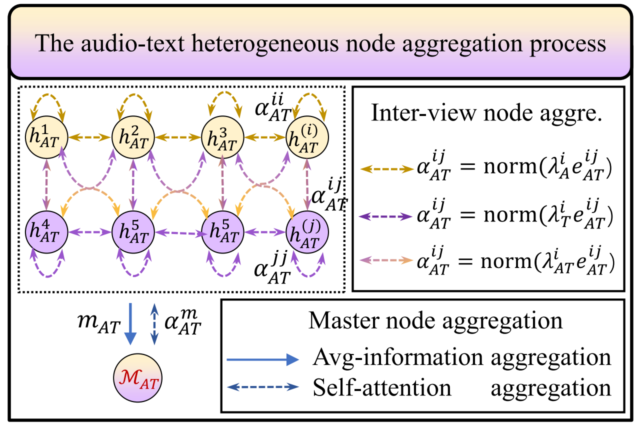


Fig. 4. Audio-text (AT) correlation measurement via the HGFM, which applies a dual-node type aggregation operation enhanced with a master node from a global view.

Based on the above observations, the set of defined inter-view relationships can be represented as $c \in \{\{A,T\},\{A,E\},\{E,T\}\}$. Specifically, the HGFM effectively updates different types of nodes through an inter-graph attention mechanism, capturing the complex interactions among heterogeneous graphs for identifying inter-view unnatural forgery correlations. In this way, each pair of correlated graphs within a specific relationship is processed through the HGFM to derive a heterogeneous graph $\mathcal{H}_c$ and a master node $\mathcal{M}_c$. Taking the AT relationship as an example, the HGFM process for a pair of graphs $\mathcal{G}_A$ and $\mathcal{G}_T$ is shown in 4. Specifically, the heterogeneous graph $\mathcal{H}_{AT} \in \mathbb{R}^{(n_a+n_t) \times d_h}$ is initially constructed by connecting the transformed nodes with each other, where d_h is the transformed dimensionality. Here, the transformed node i, denoted as $h_{AT}^{(i)} \in \mathbb{R}^{d_h}$, is derived as follows:

$$h_{AT}^{(i)} = \begin{cases} W_A^h g_A^{(i)}, & \text{if } i \in N(\mathcal{G}_A), \\ W_T^h g_T^{(i)}, & \text{if } i \in N(\mathcal{G}_T), \end{cases} \tag{5}$$

where $N(\mathcal{G}_A)$ and $N(\mathcal{G}_T)$ denote the node sets of graphs $\mathcal{G}_A$ and $\mathcal{G}_T$, respectively. The transformation matrices $W_A^h \in \mathbb{R}^{d_a \times d_h}$ and $W_T^h \in \mathbb{R}^{d_t \times d_h}$ are applied based on different types of nodes. Considering the unique characteristics and correlations of different types of nodes, the inter-graph attention mechanism incorporates a dual-node type aggregation operation, enhanced with a master node from a global view. Specifically, in $\mathcal{H}_{AT}$, the strength of the relationship e_{AT}^{ij} between nodes i and j is first measured through an IGAM, as expressed in Eq. (1). Moreover, the master node is used to capture the global information by interacting with all the other nodes in the heterogeneous graph. To this end, the average information is initially consolidated from $\mathcal{H}_{AT}$, which is denoted as follows:

$$m_{AT} = \frac{1}{n_c} \sum_{i=1}^{n_c} h_{AT}^{(i)}, \tag{6}$$

where m_{AT} describes the average information and n_c is the number of nodes in graph $\mathcal{H}_{AT}$. In this context, the interaction between the master node and node $i \in N(\mathcal{H}_{AT})$ can be formulated as $h_{AT}^{(i)} \odot m_{AT}$. Then, in $\mathcal{H}_{AT}$, the correlations of nodes i and j are adjusted via learnable vectors in terms of their node types:

$$(e_{AT}^{ij})' = \begin{cases} \lambda_A^i e_{AT}^{ij}, & \text{if } i, j \in N(\mathcal{G}_A), \\ \lambda_T^i e_{AT}^{ij}, & \text{if } i, j \in N(\mathcal{G}_T), \\ \lambda_{AT}^i e_{AT}^{ij}, & \text{otherwise}, \end{cases} \tag{7}$$

where learnable maps $\{\lambda_A, \lambda_T\} \in \mathbb{R}^{1\times d_h}$ are applied to nodes i and j that are derived from the same node set of $N(\mathcal{G}_A)$ or $N(\mathcal{G}_T)$, respectively. In addition, when nodes i and j belong to different node sets, $\lambda_{AT} \in \mathbb{R}^{1\times d_h}$ is applied. The normalized weights α_{AT}^{ij} are calculated by applying a softmax function to $(e_{AT}^{ij})'$. Then, the node aggregation process in a heterogeneous graph is described as follows:

$$h_{AT}^{(i)'} = \sum_{j\in\mathcal{N}(h_{AT}^{(i)})} \alpha_{AT}^{ij} h_{AT}^{(j)}, \tag{8}$$

where $\mathcal{N}(h_{AT}^{(i)})$ represents the neighbour of node i, including i itself. The node i is finally updated using a dual-projection operation on $h_{AT}^{(i)'}$ and $h_{AT}^{(i)}$, the process of which was described in Eq. (4). Moreover, the master node aggregation process is achieved through a self-attention mechanism that allows the master node to attach more importance to nodes that are more relevant to identifying discrepancies between real and spoofing audio. The above process is described as follows:

$$\begin{aligned} \alpha_{AT}^m &= \text{softmax}_j \left(W_{AT}^m \cdot \left(h_{AT}^{(j)} \odot m_{AT} \right) \right), \\ m'_{AT} &= \sum_{j\in N(\mathcal{H}_{AT})} \alpha_{AT}^m h_{AT}^{(j)}, \end{aligned} \tag{9}$$

where $N(\mathcal{H}_{AT})$ represents the node set of graph $\mathcal{H}_{AT}$ and the learnable map W_{AT}^m is applied to measure the importance of the master node to the other heterogeneous nodes, resulting in normalized α_{AT}^m. m'_{AT} represents the aggregated information enhanced by the master node. Finally, the master node $\mathcal{M}_{AT}$ is derived by imposing a dual-projection operation on m'_{AT} and m_{AT}, the process of which was described in Eq. (4).

Benefiting from the above components, the multi-view correlation measurement network adeptly learns to model complex multi-view interactions, facilitating the capture of both intra-view and inter-view forgery correlation cues. Through this network, the sets of homogeneous graphs $\mathcal{G}$, heterogeneous graphs $\mathcal{H}$, and master nodes $\mathcal{M}$ are derived.

D. Multi-View Synthesized Classifier

The multi-view synthesized classifier incorporates a group-based readout scheme to preserve crucial spoofing information derived from diverse cues. Given the varying performances that these cues may exhibit in spoofing detection evaluations, complementary information from these cues can be utilized for robust detection. Suppose that the combination of the given cues $\mathcal{U} = \{\mathcal{G}, \mathcal{H}, \mathcal{M}\}$ can be divided into K groups. For instance, the spoofing cues retained in $\mathcal{G}$ are captured from the intra-view nodes, while those in $\mathcal{H}$ are derived from the inter-view nodes within defined relationships.

TABLE I

OVERVIEW OF THE EVALUATED DATASETS WITH DETAILED STATISTICS

Dataset	Partition	#bonafide	#spoof	Spoofing Attack
ASVS2019LA	Train.	2580	22800	A01-A06
	Dev.	2548	22296	A01-A06
	Eval.	7355	63882	A07-A19
ASVS2021LA	Eval.	14816	133360	L01-L07
	Prog.	1676	14788	
	Hid.	1960	14966	
ASVS2021DF	Eval.	14869	519059	D01-D09
	Prog.	5768	53557	
	Hid.	1980	16596	
ASVS2015	Dev.	3494	49875	S01-S05
	Eval.	9404	184000	S01-S10
FakeorReal	Test.	2264	2370	Unknown
FakeAVCeleb	All	10209	11357	Unknown
IntheWild	All	19963	11816	Unknown

Moreover, the master node sets $\mathcal{M}$, extracted from the HGFM, represent global forgery-sensitive cues. Consequently, they can be divided into three groups, with each group subjected to a GRS operation that integrates graph pooling, node-wise maximization, and node-wise averaging operations. Specifically, graph pooling is implemented by calculating scores that reflect the relative importance of nodes within a specific graph. Taking the k-th group as an example, where the j-th graph can be denoted as $\mathcal{U}_{k,j} \in \mathbb{R}^{n_{k,j}\times d_k}$ with $n_{k,j}$ nodes. In addition, the graphs in each group have the same dimension d_k to facilitate implementation. Then, a learnable map $W_{k,j} \in \mathbb{R}^{d_k\times 1}$ is used to map $\mathcal{U}_{k,j}$ to scores $S_{k,j} \in \mathbb{R}^{n_{k,j}\times 1}$ by a dot-product as follows:

$$S_{k,j} = \text{sigmoid}\left(\mathcal{U}_{k,j} \cdot W_{k,j}\right). \tag{10}$$

Then, top-P important nodes, whose indices are selected by scores as $P = \text{TOP}_P(\mathcal{S}_{k,j})$, are aggregated by an element-wise multiplication to retain important information as follows:

$$\mathcal{U}_{k,j}^P = \{u_{k,j}^{(p)} \odot s_{k,j}^{(p)} | p \in P\}, \tag{11}$$

where $\mathcal{U}_{k,j}^P$ is the pooled graph, $u_{k,j}^{(p)}$ and $s_{k,j}^{(p)}$ denote the p-th important node and corresponding score, respectively. Furthermore, J_k pooled graphs in k-th group are derived and then concatenated into $\mathcal{U}_k^p \in \mathbb{R}^{(J_k * P)\times d_k}$. Followed by node-wise operations, the output of GRS $O_k \in \mathbb{R}^{1\times(d_k*2)}$ can be obtained as follows:

$$O_k = [\max\left(\mathcal{U}_k^p\right) \oplus \text{avg}\left(\mathcal{U}_k^p\right)], \tag{12}$$

where $\oplus$ is the concatenation operation, $\max(\cdot)$ and $\text{avg}(\cdot)$ are functions of node-wise maximization and averaging operations, respectively. Finally, an MLP, takes $O = [O_1 \oplus O_2 \oplus \ldots \oplus O_K]$ as input, and produces the binary prediction Y_{pred}.

IV. EXPERIMENTAL RESULTS

A. Experimental Settings and Metrics

1) Dataset: To evaluate the proposed framework, we conduct experiments on several public audio spoofing datasets, where spoofing audio is generated using different text-to-speech (TTS), and voice conversion (VC) algorithms and their hybrids. The detailed statistics of the experimental datasets can be found in Table I, and their descriptions are as follows:

- Intra-dataset evaluation: The ASVspoof(ASVS) 2019 logical access (LA) dataset[1] is widely adopted for testing the performance of the audio spoofing detection methods. The training partitions are commonly used for training our framework, while the evaluation partition, containing unseen attacks, is used for an intra-dataset evaluation.
- Robust-dataset evaluation: The ASVS2021 datasets[2] includes both an LA track and a deepfake (DF) track. The LA evaluation partition is used to simulate the telephone transmission and VoIP communication aspects, while the DF evaluation partition is processed with various commercial audio codecs.
- Cross-dataset evaluation: The ASVS2015 dataset[3] is adopted to evaluate the inconsistencies in TTS and VC attacks when compared with the ASVS2019LA dataset. Moreover, the data distributions of the In-the-wild dataset,[4] the FakeorReal original dataset,[5] and the FakeAVCeleb dataset[6] significantly differ from those in the ASVS family of datasets, thereby increasing the difficulty of the detection task.

In terms of the model input, we preprocess each trail in a single-channel format with a 16-kHz sampling rate and select the first 64,600 samples through truncation or repetition.

2) Evaluation Metrics: The equal error rate (EER) and minimum tandem detection cost function (MT-DCF) are adopted as experimental metrics to evaluate the achieved detection performance. Let $P_{fa}(\theta)$ be the false alarm rate at threshold θ, which can be defined as

$$P_{fa}(\theta) = \frac{N_{spoof} > \theta}{N_{trials}}, \tag{13}$$

where N_{trials} is the total number of testing trials and N_{spoof} is the number of spoofing trials with output score $S > \theta$ produced by the detection method. Let $P_{miss}(\theta)$ be the missing rate at threshold θ, which can be defined as

$$P_{miss}(\theta) = \frac{N_{bona} \leqslant \theta}{N_{trials}}, \tag{14}$$

where N_{bona} is the number of real trials with output scores $S \leqslant \theta$ produced by the detection method. The EER is the error rate obtained by the system for a specific threshold θ_{EER} in which false-positive and false-negative rates are equal, i.e.

$$\text{EER} = P_{fa}(\theta_{EER}) = P_{miss}(\theta_{EER}). \tag{15}$$

Therefore, a lower EER value indicates better detection performance. In addition, the MT-DCF is used as an assessment of a tandem system while keeping the detection system and the automatic speaker verification system isolated from each other. It considers the subsystem cost of rejecting a real trial or accepting a spoofing trial and can be defined as

$$\text{MT-DCF} = \min_{\theta_{EER}} \left\{ \beta P_{miss}(\theta_{EER}) + P_{fa}(\theta_{EER}) \right\}, \tag{16}$$

[1]https://www.asvspoof.org/index2019.html
[2]https://www.asvspoof.org/index2021.html
[3]https://www.asvspoof.org/index2015.html
[4]https://deepfake-demo.aisec.fraunhofer.de/in_the_wild
[5]https://bil.eecs.yorku.ca/datasets/
[6]https://sites.google.com/view/fakeavcelebdash-lab/

TABLE II
OVERVIEW OF THE COMPARED METHODS WITH DETAILED DESCRIPTIONS

Methods	Model Input	Feature View		
		Audio	Emotion	Text
C-GMM [16]	CQCC	★		
L-GMM [16]	LFCC	★		
L-LCNN [17]	LFCC			
E-DNN [45]	Mel-Spec		★	
R-Net2 [36]	Raw Waveform	★		
R-ST [37]	Raw Waveform	★		
A-ST, A-ST-L [38]	Raw Waveform	★		
TSSDN [35]	Raw Waveform	★		
SSLAS [43]	Raw Waveform			★
PSDL [44]	Raw Waveform			★

where β is the penalty coefficient. As with the EER, a smaller MT-DCF value indicates better detection performance.

3) Compared Methods: The details of the methods compared in our study are presented in Table II. These methods including official baselines [16], [17] and various end-to-end systems [35], [36], [37], [38], [43], [44], [45], which are recognized as outstanding systems for audio spoofing detection. Among these, we specifically select methods that leverage emotional-view features [45] or textual-view features [43], [44], as they also use an SSL-based front-end extractor, consistent with our textual-view extraction pipeline. To align these methods with our experimental settings, we retrain the SSLAS [43] model without using a data augmentation strategy and modify E-DNN [45] by replacing its random forest classifier with a fully connected layer. For PSDL [44], we use the single-resolution, utterance-level version of the pre-trained model for our evaluations.

4) Implementation Details: The front-end feature extractors are first pre-trained on specific tasks and then fine-tuned on the ASVS2019LA training partition, along with other components of the entire framework. This training strategy ensures that these components are specifically adapted for the audio spoofing detection task, resulting in more precise feature extraction and effective integration within the overall framework. Moreover, although text and emotional labels are not directly used through the training process (i.e., not multi-task learning), we employ pre-trained models that have been trained on these labels to extract relevant features. These features are then integrated into our framework and used to measure correlations among multi-view features, effectively detecting spoofing cues. For specific reproduction of multi-view extractors [55], [56], [57], we focus on methodology that how well multi-view could be integrated collectively to mine generalized spoofing cues for detection. Therefore, we select an audio base encoder [55] drawn on R-Net2 [36], R-ST [37], A-ST, and A-ST-L [38] listed in Table II. Specifically, the audio-view features $\mathcal{F}_A \in \mathbb{R}^{29\times64}$ are derived from the "encoder1" layer of the model [55]. The textual features $\mathcal{F}_T \in \mathbb{R}^{50\times64}$ are extracted using the last hidden states of the ASR encoder from wav2vec2-XLSR-300M [52]. The textual features are also utilized in compared methods, namely SSLAS [43] and PSDL [44]. Similar to E-DNN [45], the emotional-view features are extracted precedes the last fully connected layer of the SER model [58], and it is followed

TABLE III

DETECTION PERFORMANCE COMPARISON RESULTS OBTAINED ON THE ASVS2019LA EVALUATION DATASET IN TERMS OF THE POOLED EQUAL ERROR RATE (EER (%)) AND MINIMUM TANDEM DETECTION COST FUNCTION (MT-DCF). THE EERS (%) ARE PRESENTED SEPARATELY FOR DIFFERENT ATTACK TYPES. THE AVERAGE (AMSDF) AND BEST (AMSDF †) PERFORMANCES OF OUR FRAMEWORK ARE REPORTED

Methods	A07	A08	A09	A10	A11	A12	A13	A14	A15	A16	A17	A18	A19	EER	MT-DCF
C-GMM	**0.00**	**0.04**	0.14	15.16	**0.08**	4.74	26.15	10.85	1.26	**0.00**	19.62	3.81	**0.04**	9.57	0.2366
L-GMM	12.86	0.37	**0.00**	18.97	0.12	4.92	9.57	1.22	2.22	6.31	7.71	3.58	13.94	8.09	0.2116
E-DNN	0.22	3.42	0.11	0.45	0.33	0.19	0.19	0.42	0.38	0.75	20.51	17.20	6.24	7.14	0.1691
R-ST	0.34	0.53	0.08	0.38	0.29	0.33	0.30	0.29	0.27	0.83	1.71	3.11	0.73	1.07	0.0339
A-ST	0.47	0.41	**0.00**	0.79	0.16	0.67	0.12	0.16	0.51	0.65	1.28	2.65	0.67	0.82	0.0272
A-ST-L	0.61	0.20	**0.00**	1.06	0.16	0.77	0.20	0.06	0.61	0.65	1.91	3.03	0.69	0.99	0.0317
TSSDN	1.43	0.75	0.02	1.75	0.06	0.18	0.06	0.11	2.05	1.25	6.01	1.14	1.44	1.62	0.0474
SSLAS	0.06	0.06	0.02	0.40	0.10	**0.14**	**0.00**	0.06	0.24	0.06	0.37	0.84	0.35	0.25	0.0071
PSDL	0.69	1.04	0.76	1.79	0.76	0.76	0.68	1.69	2.28	0.71	0.81	1.85	4.78	1.67	0.0537
AMSDF†	0.03	0.05	0.03	0.52	0.38	0.23	0.02	0.05	0.33	0.12	0.30	0.47	0.41	0.31	0.0097
AMSDF	0.02	**0.04**	0.02	**0.23**	0.12	0.16	0.02	**0.02**	**0.16**	0.06	**0.26**	**0.29**	0.31	**0.16**	**0.0055**

by the pooling operation to yield $\mathcal{F}_E \in \mathbb{R}^{25\times 64}$. For the multi-view correlation measurement network, the node dimensionality processed in the IGAM is 64, and the temperature is 2. The number of implemented HGFM layers is set to 2, and the dimensionality of the transformed nodes d_t is set to 64. For the multi-view systhesized classifier, the considered number of groups K is 3. The graph pooling operation selects the highest 50% of the nodes for each graph. The last MLP comprises two fully connected layers, with output dimensions of 64 and 2. In line with other studies [36], [37], [38], the Adam optimizer and cross-entropy loss are used to optimize the proposed framework, and the network weights are initialized with three random seeds. All the experiments are conducted on the same platform with a single NVIDIA GeForce RTX 3090 GPU.

B. Intra-Dataset Evaluation

To investigate the detection performance of the proposed method in the intra-dataset evaluation, the evaluation partition of the ASVS2019LA dataset is utilized. The results are shown in Table III, from which we can find the following.

1) Hand-Crafted Audio Features: The high EERs produced by hand-crafted feature-based detection methods (namely, C-GMM, L-GMM, and E-DNN) indicate a potential limitation in their ability to effectively detect realistic-sounding audio synthesized by unseen and advanced deepfake algorithms. This limitation may occur because these methods focus mainly on specific forgery artifacts within the audio spectrum. For example, the Mel-spectrogram feature in E-DNN emphasizes lower audio frequencies, the CQCC feature in C-GMM captures spoofing cues within the lower frequency bands, and the LFCC feature in L-GMM provides a balanced representation across the entire frequency spectrum. However, this specialized focus might lead to a less comprehensive analysis of spoofing characteristics, thereby compromising the detection performance.

2) Learned Audio-View Features: These end-to-end methods achieve satisfactory performances by directly learning audio-view spoofing cues from raw waveform, e.g., A-ST with an EER of 0.82%. This indicates that the audio-view features learned through a well-designed network are sufficiently powerful for tackling the audio spoofing detection task in an intra-dataset evaluation.

3) Other Single-View Features: The results show that the use of emotional-view features with a DNN-based classifier layer (namely E-DNN), provides a strong discrimination ability for detecting specific spoofing attacks. This is achieved by estimating the intensity or quantity of the expressed emotion in the given audio, as demonstrated in work [45]. However, when faced with a very efficient VC spoofing system, especially ranging from A17 to A19, the performance of emotional features may not be as well. In addition, the textual-view features with a sophisticated decoder (namely, SSLAS or PSDL), exhibit varying performances in the intra-dataset evaluation. Specifically, SSLAS achieves an EER of 0.25% and an MT-DCF of 0.007, indicating the best detection performance among the compared methods. This superior performance may be attributed to the extensive pre-training of this feature extractor on diverse corpora.

Moreover, our proposed framework achieves state-of-the-art performance with an EER of 0.16% and an MT-DCF of 0.006 in the intra-dataset evaluation. A possible reason for this improvement is that the above methods, including SSLAS, utilize only single-view features to capture forgery artifacts, whereas our framework absorbs multi-view information to achieve better detection effects.

C. Robust-Dataset Evaluation

To evaluate the robustness of several detection methods in realistic compression scenarios, the ASVS2021LA dataset and the ASVS2021DF dataset are adopted. In the standard evaluation setting, we assess various methods using the evaluation partitions of robust datasets, aligning with the procedures used in most related works. In this context, the specific EER of distinct compression algorithms, the pooled EER (EER_S) and the MT-DCF are reported. Furthermore, to conduct a comprehensive investigation across robust datasets, especially in non-silent scenarios, we exclude the silent part within the progress partition and the hidden track partition. Here, the EER_NS is reported. The overall results are shown in Table IV, and we can learn the following.

1) Standard Evaluation: The compression scenario presents a significant challenge for the audio spoofing detection task,

TABLE IV

ROBUST PERFORMANCE COMPARISON RESULTS OBTAINED ON THE ASVS2021 EVALUATION DATASETS IN TERMS OF THE POOLED EER(%) (EER_S) AND MT-DCF METRICS. THE EER (%) FOR EACH SPECIFIC ATTACK TYPE IS DETAILED SEPARATELY. ADDITIONALLY, THE METRIC FOR THE WHOLE ASVS2021 DATASETS THAT INCLUDE NON-SILENT AUDIO, IS REPRESENTED AS EER_{NS}

Methods	LA								DF						
	LA03	LA04	LA05	LA06	LA07	EER_S	MT-DCF	EER_{NS}	DF02	DF03	DF04	DF05	DF08	EER_S	EER_{NS}
R-Net2	16.72	6.41	6.33	10.66	7.95	9.50	0.4257	14.74	27.63	27.49	26.72	27.23	18.74	22.38	22.13
R-ST	12.43	6.45	5.94	12.69	10.02	10.17	0.4627	16.38	28.16	27.75	27.40	27.84	19.58	23.22	23.68
A-ST	13.82	7.89	6.32	9.55	12.14	10.91	0.4931	15.38	24.75	23.88	23.73	24.20	18.49	20.96	20.69
A-ST-L	19.30	7.79	6.03	11.76	11.88	12.82	0.5468	15.70	24.12	23.79	23.46	23.47	17.40	20.25	18.72
TSSDN	36.94	3.42	3.10	11.20	4.92	13.86	0.5743	17.89	36.31	35.77	35.34	35.36	24.72	29.98	24.28
SSLAS	11.76	1.50	3.15	8.61	4.37	7.36	0.3769	10.39	8.99	9.74	8.63	8.99	7.50	8.13	8.53
PSDL	10.99	3.82	3.81	7.25	16.77	8.22	0.4155	14.80	31.14	20.62	26.68	24.38	28.92	21.19	21.69
AMSDF†	5.73	1.12	1.36	7.39	2.84	3.63	0.2859	9.22	10.40	7.25	5.82	5.15	7.44	6.06	7.53
AMSDF	**2.33**	**0.72**	**0.81**	**3.81**	**1.95**	**2.00**	**0.2408**	**7.90**	**4.91**	**3.60**	**3.49**	**3.21**	**4.54**	**3.82**	**5.47**

as it may blur captured forgery cues that are critical for detection purposes. Therefore, when compared to their intra-dataset evaluation performance, the detection performance in the robust-dataset scenario exhibits varying degrees of degradation. For example, SSLAS, without using data augment, degrades from an average EER of 0.25% to 7.36% (or 8.13%) on the ASVS2021LA (or ASVS2021DF) dataset. Although the impact of compression is inevitable in realistic scenarios, this effect can be mitigated by enhancing the robustness of the detection model. One possible way to achieve improved robustness is by utilizing a data augmentation strategy. However, this paper takes the multi-view learning approach to boost robustness, which modeling high-level spoofing cues by measuring multi-view correlations. Through this approach, our framework achieves EERs of 2% and 3.59% on ASVS2021LA and ASVS2021DF, respectively, indicating a more moderate decline. This finding suggests that capturing the correlation among multi-view features can provide robust spoofing cues to detect audio spoofing patterns.

2) Non-Silent Evaluation: The research presented in [59] indicated that a training dataset contains an uneven silence distribution, which may cause the model to overly focus on the silent parts. As shown in the EER_{NS} results, the removal of silent parts from the audio is a key factor that leads to a significant performance decline. Even under such challenging conditions, e.g., detection tasks conducted on robust datasets containing non-silent audio, the EERs of our framework remain below 7.9%. However, this also highlights the shortcomings of our work: the absence of a specialized design for silent segments at the model level.

In conclusion, our framework leverages multi-view features derived from audio and applies a designed multi-view correlation measurement network to mine inconsistent correlations introduced by the spoofing attacks. With these designs, even in the challenging robust-dataset evaluation, we can effectively mitigate the detection performance degradation caused by compression and silence parts in realistic scenarios.

D. Cross-Dataset Evaluation

To evaluate the generalization performance of several detection methods, we conduct experiments on various cross datasets, including ASVS2015, FakeOrReal, FakeAVCelebac, and InTheWild. The results of the cross-dataset evaluation are shown in Table V, and the confusion matrices are presented in Fig.5, from which we can find the following.

TABLE V

GENERALIZATION PERFORMANCE ACHIEVED ON CROSS DATASETS IN TERMS OF THE POOLED EER(%) METRIC. THE AVERAGE EER IS DENOTED AS EER_{AVG}(%)

Methods	ASVS2015		FakeorReal	FakeAVCeleb	IntheWild	EER_{AVG}
	Dev.	Eval.				
R-ST	10.34	6.63	48.94	60.65	52.52	35.82
A-ST	5.69	3.22	44.26	52.93	43.50	29.92
A-ST-L	7.92	5.25	31.40	53.13	44.64	28.47
TSSDN	7.15	10.58	36.17	36.39	34.16	24.89
SSLAS	0.26	0.26	8.98	**5.25**	18.67	6.68
PSDL	6.35	4.46	23.46	19.00	32.81	17.21
AMSDF†	0.14	0.13	8.30	22.10	18.06	9.75
AMSDF	**0.06**	**0.12**	**4.55**	8.03	**9.50**	**4.45**

1) Single-View Features: These compared methods, which rely on single-view spoofing artifact cues, generally achieve good performance in the intra-dataset evaluation. However, they exhibit unsatisfactory results in the cross-dataset evaluation, particularly on datasets whose data distribution is significantly different from that of the ASVS family of datasets (especially FakeOrReal, FakeAVCelebac, and InTheWild). For example, A-ST achieves an EER of 0.82% in the intra-dataset evaluation, while its average EER decreases to 29.92% in the cross-dataset evaluation. The possible reason for this may be that these models are designed to capture specific characteristics and noise of the ASVS training dataset. The reliance on specific forgery artifacts using single-view features consequently limits the detection improvement when faced with diverse spoofing attacks.

2) Textual-View Features: The results of cross-dataset evaluation reveal that among the various multi-view features, the textual-view feature exhibits the most impressive performance. For example, SSLAS achieves an average EER of 6.68%. This outcome further demonstrates that textual-view spoofing cues, when learned from a broad and realistic corpus, effectively help the utilized model counter unknown audio spoofing algorithms. Moreover, by leveraging multi-view information to model spoofing correlation cues, our framework achieves better generalization performance with an average EER of 4.45%. It is worth noting that the SSLAS outperforms our framework on the FakeAVCeleb dataset, but as depicted in Fig.5(c), it produces a higher false-negative rate (FNR)

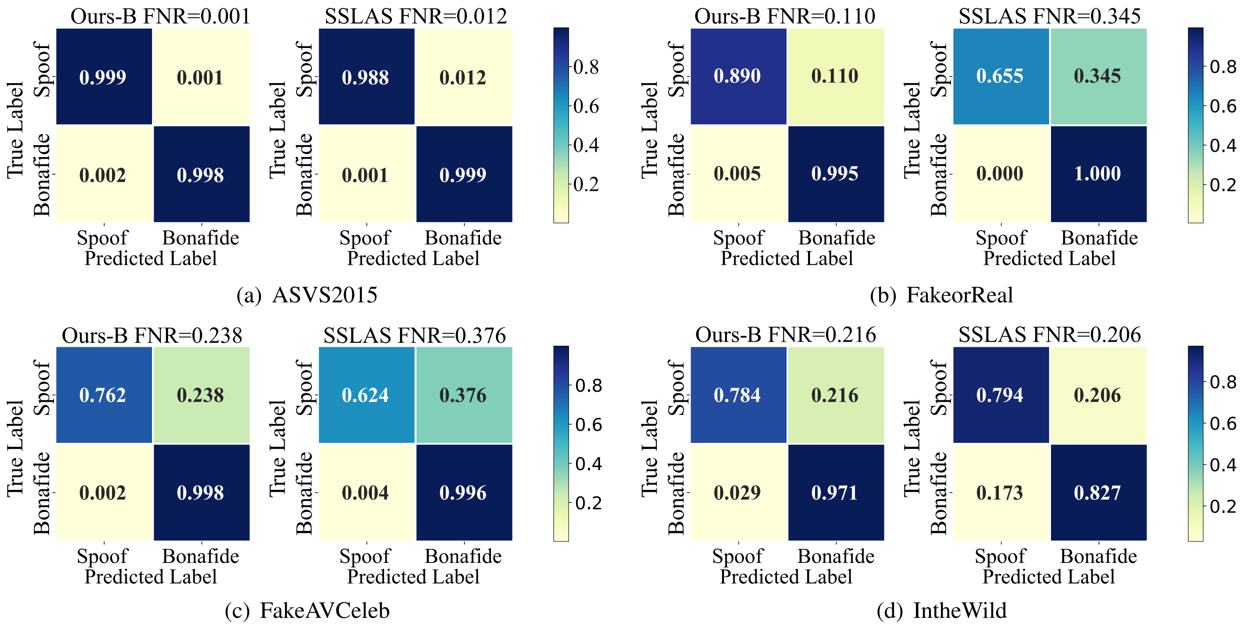


Fig. 5. Confusion matrices obtained by SSLAS and AMSDF† on the cross-dataset evaluations. Lower FPR or FNR achieved by our framework indicates a better detection performance without overfitting to real or spoofing patterns.

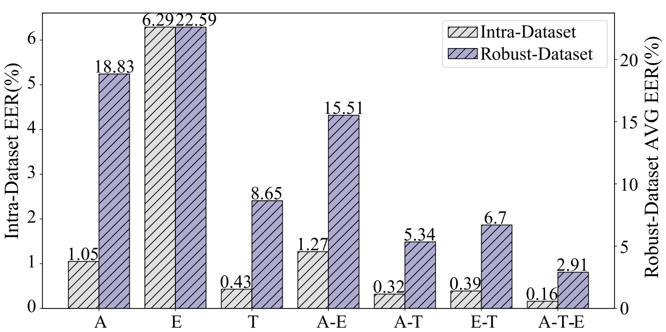


Fig. 6. Comparison among the multi-view integration strategies. The left y-axis illustrates the intra-dataset pooled EER, while the right y-axis presents the average EER attained across robust datasets.

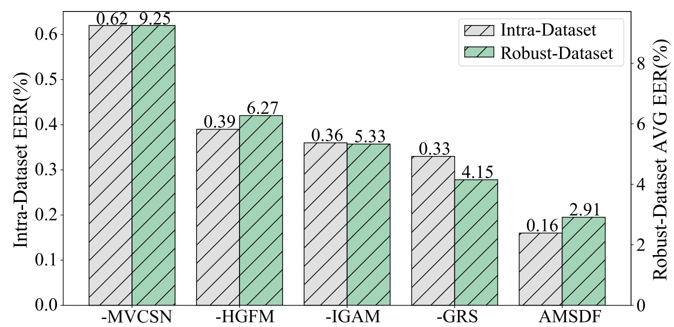


Fig. 7. Comparison among various components of our framework. The left y-axis indicates the intra-dataset pooled EER, while the right y-axis displays the average EER attained across robust datasets.

of 37.6%. This finding suggests that SSLAS might overfit spoofing audio, thereby misclassifying them as real audio. Furthermore, Fig.5 also indicates that our framework attains lower false-positive rates (FPRs) and FNRs in the cross-dataset evaluation, suggesting that it can mitigate overfitting to either spoofing or real audio. Overall, this study demonstrates a better generalization performance of our framework.

E. Ablation Study

To demonstrate the effectiveness of the multi-view integration strategies and different components of our proposed framework, ablation studies are conducted based on intra-dataset and robust-dataset scenarios. The detailed experimental analyses are as follows.

1) Comparative Analysis of the Multi-View Integration Strategies: This study aims to evaluate the contribution of the multi-view features utilized in the proposed framework. To this end, three integration strategies are adopted, including single-view features: removing HGFM, using IGAM for single-view graph extraction and then using a single GRS for classification; dual-view features: using two IGAMs and a single HGFM to measure their correlation and then using a multi-view synthesized classifier for classification; multi-view features: overall structure. As shown in Fig. 6, the results indicate the varying impacts of the different correlations among multi-view features measured on the overall performance of our framework. Specifically, under the single-view conditions, the textual-view features extracted from a well-pretrained ASR model outperform the other features, yielding average EERs of 0.43% for the intra-dataset evaluation and 8.65% for the robust-dataset evaluation. This superior performance can be attributed to the ability to absorb diverse and useful information from training corpora, resulting in more effective representations. Moreover, the performance improves when textual-view features are complemented with others. For instance, when measuring audio-textual inter-view correlations, the average EER of robust-dataset evaluation is improved by 38.27% over that produced when using single

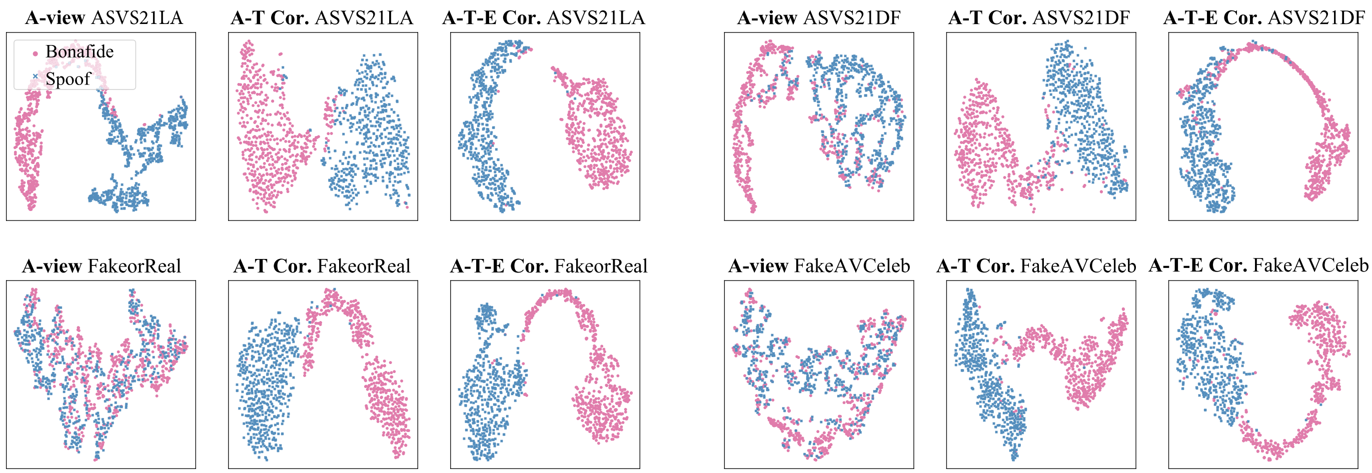


Fig. 8. t-SNE Visualizations of the features extracted under different multi-view integration strategies, including the audio-view (A), Audio-Text view (A-T), and Audio-Emotion-Text view (A-E-T). The visualizations are based on the average results of 1000 real samples (red dots) and 1000 spoofing samples (blue dots). The results on robust-dataset scenarios (ASVS2021LA, ASVS2021DF) and cross-dataset scenarios (FakeorReal, FakeAVCeleb) are presented.

textual-view features. This indicates that identifying discrepancies in the inter-view correlations can facilitate the framework to detect distorted spoofing correlation cues. Furthermore, features extracted from ASR and SER, being pre-trained on real corpora, are unseen to spoofing audio and thus can be effectively complemented by audio-view features learned from the audio spoofing dataset. In other words, integrating these three features can provide more correlation cues about real and spoofing patterns. Consequently, the best performance is achieved by fully measuring correlations among multi-view features, i.e. adopting AET within the proposed framework.

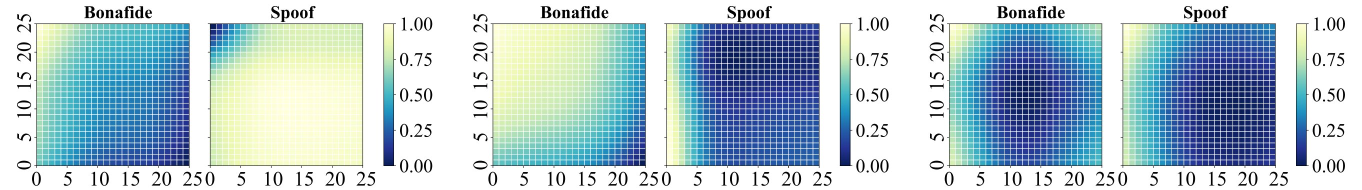
2) Comparative Analysis of the Various Components in the Proposed Framework: In this study, we remove or replace different components with an MLP-based classifier in the proposed framework to show their individual effects. As shown in Fig. 7, the results indicate the critical role of the multi-view correlation measurement network (MVCSN) in providing robust and generalization spoofing cues for our framework. Specifically, when comparing the performances achieved with and without this network, the average EER changes from 0.16% to 0.62% in the intra-dataset evaluation, and shifts from 2.91% to 9.25% in the cross-dataset evaluation. Additionally, after removing the HGFM from the multi-view correlation measurement network, we observe a similar shift in the average EER during the cross-dataset evaluation, from 2.91% to 6.27%. This significant EER difference underscores the critical role of measuring inter-view correlations for distinguishing between real and spoofing audio. Similarly, when we remove the IGAM from the multi-view correlation measurement network, the resulting degradation in the average EER reaffirms its importance for effectively capturing intra-view forgery correlation cues. Furthermore, employing the GRS before the MLP plays a crucial role in integrating diverse spoofing cues, thus significantly enhancing the detection performance of the proposed framework. In conclusion, each component of our proposed framework serves a significant purpose and is thoughtfully designed, contributing to the overall effectiveness shown in the audio spoofing detection task.

F. Model Visualization

To evaluate the effectiveness of our multi-view learning approach in robust and cross-dataset evaluations, we employ t-SNE [60] to visualize the features extracted from different view integrations, including text-only (T), audio-text (AT), and audio-emotion-text (AET) cases. Furthermore, to investigate the inconsistent correlations among the specific relationships between spoofing and real audio, we visualize the edge weights in a heterogeneous graph.

1) t-SNE Visualization: The t-SNE visualization results, as shown in Fig. 8, reveal the distinct clustering patterns and distributions of each dataset under different feature integrations. Specifically, in the robust datasets, when using only the audio-view features, some real audio (represented as red dots) are misclassified as spoofing audio (represented as blue dots), potentially due to compression progress obscuring the artificial forgery traces. When confronted with cross-dataset scenarios, the clustering performance deteriorates, as evidenced by the intermingling of red and blue dots. However, this situation becomes mitigated with the addition of the textual-view features (AT) as the distributions of spoofing and real audio begin to disperse. Furthermore, the separation of real and spoofing audio through multi-view (AET) features integration is markedly clearer, with only a few dots intermixed at the boundary. Overall, measuring correlations among multi-view features benefits the robustness and generalization performance of our framework, which is in alignment with our ablation studies.

2) Edge Weight Visualization: As observed in Fig. 9 (a), it is interesting to note that the AT correlation captured by the HGFM is stronger (brighter in the figure) for spoofing patterns than for real patterns. This may be attributed to the fact that speech-synthesized generators are designed to produce speech that closely follows a textual script, leading to a high degree of alignment between the audio and textual features. Consequently, this finding indicates that the existing generators still fall short in terms of modeling the natural variations that are present in real human speech. Moreover, as depicted in Fig. 9(b) and 9(c), the AE correlation difference between



(a) Audio-Text (AT) correlation (b) Emotion-Text (ET) correlation (c) Audio-Emotion (AE) correlation

Fig. 9. Visualizations of the edge weights in heterogeneous graphs, indicating inter-view correlations measured by the HGFM. The visualizations are based on the average results of 10000 real samples (red dots) and 10000 spoofing samples (blue dots). The brighter depicted in the heatmap indicates stronger correlations between corresponding inter-view nodes.

real and spoofing audio is weaker than the ET correlation difference. It is possible that while synthetic speech algorithms can replicate basic emotional elements in speech, they do not yet match the naturalness required to fully align these elements with the corresponding textual context. The above analyses also indicate that the presence of textual-view features acts as a bridge to complement other features, thereby facilitating the HGFM in effectively discerning correlation differences between real and spoofing audio. Overall, these findings align with the conclusions drawn from the experimental studies, underscoring the importance of multi-view correlation measurement in our framework.

V. Conclusion

In this study, we propose a novel audio multi-view spoofing detection framework named AMSDF. Compared to methods using domain generalization or data augmentation, our framework solely focuses on traditional training strategy but provides a new solution to capture spoofing-introduced correlation cues for detection. The components of our framework are threefold. First, multi-view feature extractors are employed to acquire comprehensive audio information for correlation analysis, including waveform features, textual representations, and emotional semantics. Second, a multi-view correlation measurement network is designed to measure the correlations among multi-view features for capturing both inter-view and intra-view forgery correlation cues. Finally, a multi-view synthesized classifier is introduced to efficiently integrate forgery cues captured from various feature sets. Benefiting from the designed components, the proposed framework can mine generalized spoofing cues as it measures unnatural correlations inherent in spoofing patterns instead of capturing specific forgery artifacts. However, a limitation of our approach is evident in the non-silence scenarios, where a lack of a specialized design within our framework can lead to overfitting to audio silent parts. In addition, the extraction of textual features is time-consuming, adversely affecting the inference speed. Therefore, our further work will explore different training strategies to develop a lightweight and generalized framework for addressing more challenging scenarios, such as silence attacks.

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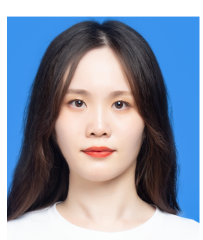

Junyan Wu received the B.S. degree from Shandong Normal University in 2019 and the M.S. degree from National Huaqiao University in 2022. She is currently pursuing the Ph.D. degree with the School of Computer Science and Engineering, Sun Yat-sen University, Guangzhou, China. Her research interests include multimedia forensics and security.

Qilin Yin received the B.S. and M.S. degrees from Nanjing University of Information Science and Technology in 2018 and 2021, respectively. He is currently pursuing the Ph.D. degree with the School of Computer Science and Engineering, Sun Yat-sen University, Guangzhou, China. His research interests include multimedia forensics and security.

Ziqi Sheng received the B.S. degree in network engineering from Jiangxi University of Science and Technology, China, in 2017, and the M.S. degree in computer science and technology from Sun Yat-sen University, Guangzhou, China, in 2021, where she is currently pursuing the Ph.D. degree in cyberspace security. Her research interests include multimedia security and forensics.

Wei Lu (Member, IEEE) received the B.S. degree in automation from Northeast University, China, in 2002, and the M.S. and Ph.D. degrees in computer science from Shanghai Jiao Tong University, China, in 2005 and 2007, respectively. He was a Research Assistant with The Hong Kong Polytechnic University from 2006 to 2007. He is currently a Professor with the School of Computer Science and Engineering, Sun Yat-sen University, Guangzhou, China. His research interests include multimedia forensics and security, and AI security.

Jiwu Huang (Fellow, IEEE) received the B.S. degree from Xidian University, Xi'an, China, in 1982, the M.S. degree from Tsinghua University, Beijing, China, in 1987, and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences, Beijing, in 1998. He is currently a Professor with the Faculty of Engineering, Shenzhen MSU-BIT University, Shenzhen, China. He has coauthored more than 350 papers in important journals and conferences, with Google citations more than 21000 and an H-index of 76. His current research interests include multimedia forensics and security.

Bin Li (Senior Member, IEEE) received the B.E. degree in communication engineering and the Ph.D. degree in communication and information system from Sun Yat-sen University, Guangzhou, China, in 2004 and 2009, respectively. From 2007 to 2008, he was a Visiting Scholar with New Jersey Institute of Technology, Newark, NJ, USA. In 2009, he joined Shenzhen University, Shenzhen, China, where he is currently a Professor. He is also the Director of Guangdong Key Laboratory of Intelligent Information Processing and Shenzhen Key Laboratory of Media Security. His current research interests include multimedia forensics, image processing, and deep machine learning. He is an Associate Editor of IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY.